Trigonometry: sin(x), cos(x), tan(x) in Degrees

Questions 1–54

Question 1. Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = \sin x$

b) $y = \cos x$

c) $y = \tan x$

Question 2. Solve the following equations:

a) $\sin\theta = 0$, $0° \leq \theta < 360°$

b) $\sin\theta = 1$, $0° \leq \theta < 360°$

c) $\sin\theta = -1$, $0° \leq \theta < 360°$

Question 3. Solve the following equations:

a) $\cos\theta = 0$, $0° \leq \theta < 360°$

b) $\cos\theta = 1$, $0° \leq \theta < 360°$

c) $\cos\theta = -1$, $0° \leq \theta < 360°$

Question 4. Solve the following equations:

a) $\tan\theta = 0$, $0° \leq \theta < 360°$

b) $\tan\theta = 1$, $0° \leq \theta < 360°$

c) $\tan\theta = -1$, $0° \leq \theta < 360°$

Question 5. Find the values of x for which $y = \sin x$ is an increasing function, for $0° \leq x \leq 360°$.

Question 6. Find the values of x for which $y = \cos x$ is a decreasing function, for $0° \leq x \leq 360°$.

Question 7. Solve the following equations:

a) $\sin\theta = \frac{1}{3}$, $0° \leq \theta < 360°$

b) $\cos\theta = \frac{2}{3}$, $0° \leq \theta < 360°$

c) $\tan\theta = \frac{4}{3}$, $0° \leq \theta < 360°$

Question 8. Solve the following equations:

a) $\sin\theta = 0.4$, $-360° \leq \theta < 0°$

b) $\cos\theta = \frac{3}{7}$, $-360° \leq \theta < 0°$

c) $\tan\theta = \frac{4}{9}$, $-360° \leq \theta < 0°$

Question 9. Solve the following equations:

a) $\sin\theta = -0.2$, $0° \leq \theta < 360°$

b) $\cos\theta = -\frac{4}{7}$, $0° \leq \theta < 360°$

c) $\tan\theta = -4$, $0° \leq \theta < 360°$

Question 10. Solve the following equations:

a) $\sin\theta = -0.9$, $-360° \leq \theta < 0°$

b) $\cos\theta = -\frac{3}{5}$, $-360° \leq \theta < 0°$

c) $\tan\theta = -\frac{9}{2}$, $-360° \leq \theta < 0°$

Question 11. Solve the following equations:

a) $\sin\theta = \frac{3}{8}$, $-360° \leq \theta < 360°$

b) $\cos\theta = 0.7$, $-360° \leq \theta < 360°$

c) $\tan\theta = 6$, $-360° \leq \theta < 360°$

d) $\sin\theta = -\frac{6}{7}$, $-360° \leq \theta < 360°$

e) $\cos\theta = -\frac{2}{9}$, $-360° \leq \theta < 360°$

f) $\tan\theta = -1.3$, $-360° \leq \theta < 360°$

Question 12. Find the values of x for which $y = \sin x$ is a decreasing function, for $-720° \leq x \leq -360°$.

Question 13. Find the values of x for which $y = \cos x$ is an increasing function, for $180° \leq x \leq 540°$.

Question 14. Solve the following equations:

a) $\sin\theta = \frac{4}{7}$, $0° \leq \theta < 720°$

b) $\cos\theta = -\frac{8}{9}$, $720° \leq \theta < 1080°$

c) $\tan\theta = -9$, $-720° \leq \theta < -540°$

Question 15. Solve the following equations:

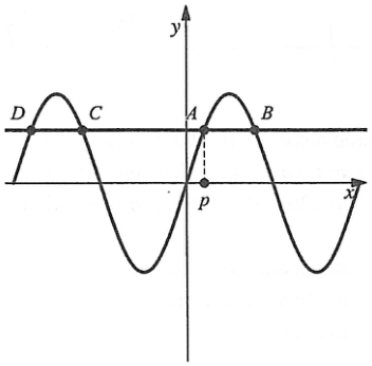
a) $\frac{1}{\sin\theta} = 18$ for $0° \leq \theta < 180°$

b) $\frac{5}{\tan\theta} = 13$ for $180° \leq \theta < 540°$

c) $\frac{0.1}{\cos\theta} = 2$ for $-180° \leq \theta < 180°$

Question 16. A student is solving a problem with a triangle. They correctly use the sine rule to find $\frac{\sin A}{18} = \frac{\sin 20°}{15}$. Find the two possible values of A.

Question 17. The graph below shows the curve $y = \sin x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.

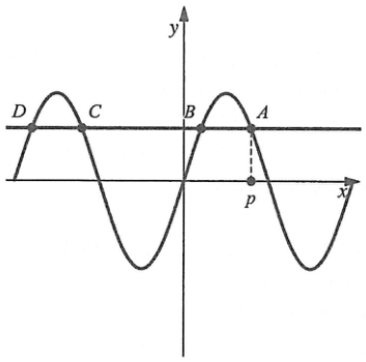


Question 18. It is given that $\sin 60° = \frac{\sqrt{3}}{2}$. Explain why $\sin 480° = \frac{\sqrt{3}}{2}$.

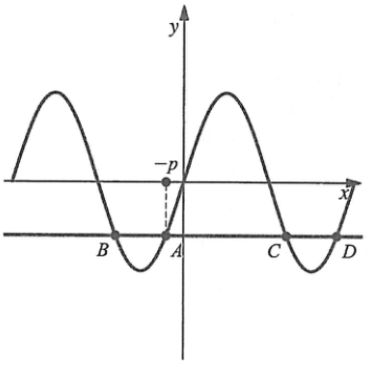
Question 19. It is given that $\cos 45° = \frac{\sqrt{2}}{2}$. Explain why $\cos 495° = -\frac{\sqrt{2}}{2}$.

Question 20. It is given that $\tan(-210°) = -\frac{\sqrt{3}}{3}$. Explain why $\tan 390° = \frac{\sqrt{3}}{3}$.

Question 21. The graph below shows the curve $y = \sin x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 22. The graph below shows the curve $y = \sin x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $-1 < k < 0$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 23. Solve the following equations:

a) $\cos^2\theta = \frac{1}{16}$ for $0° \leq \theta < 360°$

b) $3\tan^2\theta = 0.8$ for $-180° \leq \theta < 180°$

c) $10\sin^2\theta = 7$ for $0° \leq \theta < 540°$

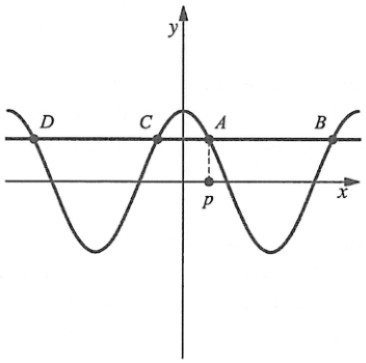
Question 24. Solve the following equations:

a) $\cos^2\theta = 2\cos\theta$ for $0° \leq \theta < 360°$

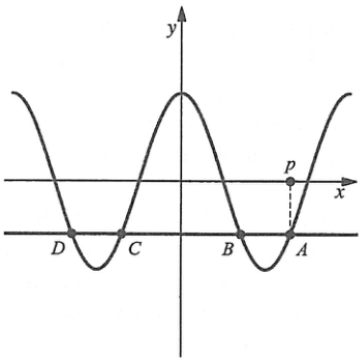
b) $7\tan^2\theta + 4\tan\theta = 0$ for $0° \leq \theta < 360°$

c) $2\sin\theta = 5\sin^2\theta$ for $0° \leq \theta < 360°$

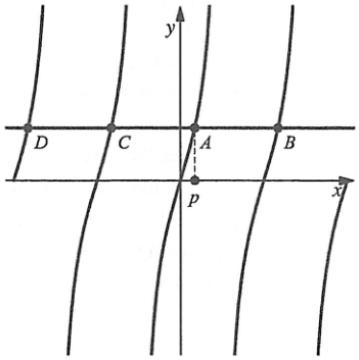
Question 25. The graph below shows the curve $y = \cos x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 26. The graph below shows the curve $y = \cos x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $-1 < k < 0$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 27. The graph below shows the curve $y = \tan x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



Question 28. Solve the following equations:

a) $3\sin\theta\cos\theta = \cos\theta$ for $0° \leq \theta < 360°$

b) $8\sin\theta\tan\theta - 9\sin\theta = 0$ for $0° \leq \theta < 360°$

c) $6\cos\theta - 5\sin\theta\cos\theta = 0$ for $0° \leq \theta < 360°$

Question 29. Solve the following equations:

a) $2\sin^2\theta + \sin\theta - 1 = 0$ for $0° \leq \theta < 360°$

b) $3\cos^2\theta - \cos\theta - 2 = 0$ for $0° \leq \theta < 360°$

c) $4\tan^2\theta + 5\tan\theta - 6 = 0$ for $0° \leq \theta < 360°$

Question 30. Solve the equations for $0° \leq \theta < 360°$:

a) $6\sin\theta = 1 + \frac{1}{\sin\theta}$

b) $9\cos\theta = \frac{16}{\cos\theta} - 32$

c) $30\tan\theta = 23 + \frac{40}{\tan\theta}$

Question 31. Prerequisite knowledge for Q31: Quadratics It is given that $f(x) = 3x^2 - 12x + 17$.

a) Write $f(x)$ in the form $a(x + b)^2 + c$, where a, b and c are integers.

b) Write down the minimum point of the curve with equation $y = f(x)$.

It is given that $g(\theta) = 3\sin^2\theta - 12\sin\theta + 17$.

c) Using your answers to a) and b), find the minimum value of $g(\theta)$ and then the smallest positive value of θ, in degrees, for which this minimum occurs.

Question 32. Write down the period of the following functions in degrees:

a) $y = \sin x$

b) $y = \cos x$

c) $y = \tan x$

d) $y = \sin 2x$

e) $y = \cos 3x$

f) $y = \tan 4x$

g) $y = \sin \frac{1}{3}x$

h) $y = \cos \frac{2}{3}x$

i) $y = \tan \frac{4}{5}x$

Question 33. State the number of solutions to the following equations. You are not required to solve them.

a) $\sin 2x = 0.3$ for $0° \leq x < 360°$

b) $\sin 2x = 0.3$ for $0° \leq x < 180°$

c) $\sin 2x = 0.3$ for $0° \leq x < 720°$

Question 34. State the number of solutions to the following equations. You are not required to solve them.

a) $\cos 2x = 0.8$ for $0° \leq x < 180°$

b) $\cos 3x = 0.8$ for $0° \leq x < 360°$

c) $\cos 4x = 0.8$ for $0° \leq x < 720°$

Question 35. State the number of solutions to the following equations. You are not required to solve them.

a) $\tan 2x = 0.8$ for $0° \leq x < 180°$
b) $\tan 3x = 0.8$ for $0° \leq x < 720°$
c) $\tan 4x = 0.8$ for $-270° \leq x < 360°$

Question 36. Write down the period of the following functions in degrees:

a) $y = \sin(x - 15°)$
b) $y = \cos(x + 80°)$
c) $y = \tan(x - 300°)$
d) $y = \sin(60° - x)$
e) $y = \cos(45° - x)$
f) $y = \tan(5° - x)$
g) $y = \sin(2x + 50°)$
h) $y = \cos(4x - 90°)$
i) $y = \tan(9x + 35°)$
j) $y = \sin(100° - \frac{1}{5}x)$
k) $y = \cos(\frac{315° - x}{8})$
l) $y = \tan(\frac{28° - 5x}{9})$

Question 37. Solve the following equations:

a) $\sin(\theta + 10°) = 0.3$ for $0° \leq \theta < 360°$

b) $\sin(\theta - 50°) = \frac{2}{11}$ for $0° \leq \theta < 360°$

c) $\sin(\theta + 205°) = -0.4$ for $0° \leq \theta < 360°$

Question 38. Solve the following equations:

a) $\cos(\theta - 60°) = \frac{4}{11}$ for $0° \leq \theta < 360°$

b) $\cos(\theta + 25°) = 0.85$ for $0° \leq \theta < 360°$

c) $\cos(\theta - 195°) = -\frac{7}{12}$ for $0° \leq \theta < 360°$

Question 39. Solve the following equations:

a) $\tan(\theta - 45°) = 2.5$ for $0° \leq \theta < 360°$

b) $\tan(\theta + 80°) = \frac{5}{6}$ for $0° \leq \theta < 360°$

c) $\tan(\theta - 215°) = -4.2$ for $0° \leq \theta < 360°$

Question 40. Find the values of x for which $y = \sin(x - 30°)$ is an increasing function, for $0° \leq x \leq 360°$.

Question 41. Solve the following equations:

a) $\sin 2\theta = \frac{\sqrt{2}}{2}$ for $0° \leq \theta < 360°$

b) $\cos 2\theta = \frac{\sqrt{2}}{4}$ for $0° \leq \theta < 360°$

c) $\tan 2\theta = \frac{\sqrt{3}}{3}$ for $0° \leq \theta < 360°$

Question 42. Solve the following equations:

a) $\sin 3\theta = 0.8$ for $0° \leq \theta < 360°$

b) $\cos 3\theta = -0.3$ for $0° \leq \theta < 360°$

c) $\tan 3\theta = -3$ for $0° \leq \theta < 360°$

Question 43. Solve the following equations:

a) $\sin \frac{1}{2}\theta = 0.2$ for $0° \leq \theta < 720°$

b) $\cos \frac{1}{4}\theta = -\frac{4}{5}$ for $0° \leq \theta < 3600°$

c) $\tan \frac{2}{3}\theta = -5$ for $0° \leq \theta < 270°$

Question 44. Solve the following equations:

a) $\sin(2\theta - 25°) = \frac{1}{2}$ for $0° \leq \theta < 360°$

b) $\cos(3\theta + 50°) = \frac{1}{2}$ for $0° \leq \theta < 360°$

c) $\tan(2\theta + 65°) = 1$ for $0° \leq \theta < 360°$

Question 45. Solve the following equations:

a) $\sin(4\theta + 161°) = \frac{5}{6}$ for $0° \leq \theta < 180°$

b) $\cos(6\theta - 94°) = 0.03$ for $-90° \leq \theta < 90°$

c) $\tan(5\theta - 213°) = 8$ for $100° \leq \theta < 200°$

Question 46. Find the values of x for which $y = \cos 2x$ is a decreasing function, for $0° \leq x \leq 180°$.

Question 47. Find the values of x for which $y = \sin 3x$ is an increasing function, for $0° \leq x \leq 360°$.

Question 48. Solve the following equations:

a) $\sin(84° - 3\theta) = \frac{1}{2}$ for $0° \leq \theta < 360°$

b) $\cos(115° - 2\theta) = \frac{1}{2}$ for $0° \leq \theta < 360°$

c) $\tan(5° - 4\theta) = 1$ for $0° \leq \theta < 360°$

Question 49. Solve the following equations:

a) $\sin(28° - 5\theta) = 0.1$ for $0° \leq \theta < 144°$

b) $\cos(77° - 3\theta) = 0.99$ for $0° \leq \theta < 300°$

c) $\tan(174° - 8\theta) = -0.56$ for $0° \leq \theta < 55°$

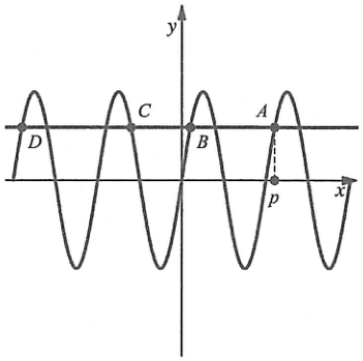
Question 50. Solve the following equations:

a) $\sin^2 2\theta - \frac{1}{4} = 0$ for $0° \leq \theta < 360°$

b) $\cos^2 3\theta - \frac{1}{16} = 0$ for $0° \leq \theta < 180°$

c) $\tan^2 5\theta - 121 = 0$ for $-40° \leq \theta < 40°$

Question 51. The graph below shows the curve $y = \sin 2x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate p. Write down the x-coordinates of B, C and D in terms of p.



Question 52. Solve the following equations:

a) $\sin(\theta - 40°) = \sin 80°$ for $0° \leq \theta < 360°$

b) $\sin(\theta + 35°) = \sin 280°$ for $0° \leq \theta < 360°$

c) $\cos(\theta + 15°) = \cos 50°$ for $0° \leq \theta < 360°$

d) $\tan(\theta - 70°) = \tan 20°$ for $0° \leq \theta < 360°$

Question 53. Solve the following equations:

a) $\sin(35° - \theta) = \sin 75°$ for $0° \leq \theta < 360°$

b) $\cos(2\theta + 80°) = \cos 55°$ for $0° \leq \theta < 360°$

c) $\tan(20° - 3\theta) = \tan 140°$ for $0° \leq \theta < 360°$

Question 54. Prerequisite knowledge for Q54: Polynomials It is given that $f(x) = 45x^3 - 63x^2 - 2x + 8$.

a) Show that $(5x - 2)$ is a factor of $f(x)$

b) Hence solve the equation $45\cos^3\theta - 63\cos^2\theta - 2\cos\theta + 8 = 0$ for $0° \leq \theta \leq 360°$.